

Pearson Edexcel International GCSE

Mathematics A

Modular Unit 1

June 2025 Reconstructed Past Paper

STUDENT WORKBOOK

Adaptive working-space edition

28 working units 94 marks

Selected from Linear Higher Papers 1H and 2H

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L-to-M | CONVERTED MODULAR EDITION

Selected from Linear Higher papers; not an original Modular examination paper.

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About this reconstructed paper

Each page contains one main demand or one small inseparable chain. The source question is followed by adaptive working space; writable grids, trees and Venn diagrams count as part of that working space.

Unit ownership rule: a question is assigned to Unit 2 when its assessed or terminal statement belongs to Unit 2, even if it uses Unit 1 prerequisites. This book contains only the demands owned by Unit 1. The 200 marks of the two Linear Higher papers were partitioned exactly; this Unit 1 selection contains 94 marks.

This is an Elite reconstructed teaching paper selected from official Linear Higher material; it is not an official Pearson Modular examination paper. Source assets are retained for private teaching use.

Contents

Unit	Source	Focus	Marks	Page
1	4MA1-1H Q2	Bounds from stated accuracy	2	2
2	4MA1-1H Q4	Index laws: multiply and divide	2	3
3	4MA1-1H Q4	Power of a product	2	4
4	4MA1-1H Q6	Solve a fractional linear equation	3	5
5	4MA1-1H Q6	Factorise, then solve	3	6
6	4MA1-1H Q9	Area of a symmetric trapezium	5	7
7	4MA1-1H Q10	Complete and draw a cubic graph	3	8
8	4MA1-1H Q11	Area of a sector	2	9
9	4MA1-1H Q12	Add algebraic fractions	3	10
10	4MA1-1H Q14	Complete a probability tree	4	11
11	4MA1-1H Q15	Powers and rationalising a surd	4	12
12	4MA1-1H Q17	Estimate a proportion from a histogram	3	13
13	4MA1-1H Q20	Cosine rule followed by sine rule	5	14
14	4MA1-1H Q21	Exactly two counters of the same colour	3	15
15	4MA1-1H Q25	Equation of the other diagonal of a square	5	16
16	4MA1-2H Q3	Subtract mixed fractions	3	17
17	4MA1-2H Q10	Connected right-angled triangles	5	18
18	4MA1-2H Q11	Recover a population from a probability	3	19
19	4MA1-2H Q12	Expand three algebraic factors	3	20
20	4MA1-2H Q14	Recognise graphs from equations	2	21
21	4MA1-2H Q15	Negative power of an algebraic fraction	3	22
22	4MA1-2H Q18	Convert a recurring decimal to a fraction	2	23
23	4MA1-2H Q20	Three-set Venn diagram and conditional probability	6	24
24	4MA1-2H Q22	Bounds for a shaded area	4	25
25	4MA1-2H Q23	Simplify a division of algebraic fractions	4	26
26	4MA1-2H Q24	Three-dimensional trigonometry in a pyramid	4	27
27	4MA1-2H Q25	Complete the square	3	28
28	4MA1-2H Q25	Sketch a quadratic from completed-square form	3	29

Working Unit 01

4MA1-1H Q2

UNIT 1**2 marks**

The height of each cup is 9 cm, correct to the nearest cm

(b) Write down the lower bound of the height.

..... cm
(1)

The weight of each cup is 120 g, correct to the nearest 10 g

(c) Write down the upper bound of the weight.

..... g
(1)

WORKING SPACE

Working Unit 02

4MA1-1H Q4

UNIT 1**2 marks**

4 $x^5 \times x^7 = x^m$

(a) Find the value of m

$m = \dots\dots\dots$
(1)

$y^8 \div y^3 = y^n$

(b) Find the value of n

$n = \dots\dots\dots$
(1)

WORKING SPACE

Working Unit 03

4MA1-1H Q4

UNIT 1

2 marks

(c) Simplify fully $(5a^4r^2)^3$

(2)

WORKING SPACE

Working Unit 04

4MA1-1H Q6

UNIT 1**3 marks**

6 (a) Solve $x - 4 = \frac{3 + 2x}{6}$

Show clear algebraic working.

$x = \dots\dots\dots$
(3)

WORKING SPACE

Working Unit 05

4MA1-1H Q6

UNIT 1**3 marks**

(b) (i) Factorise $y^2 - 11y + 30$

.....
(2)

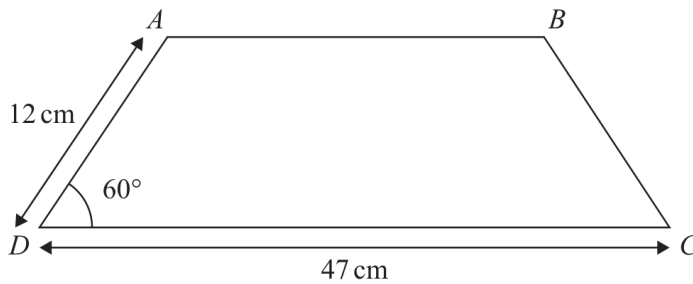
(ii) Hence solve $y^2 - 11y + 30 = 0$

.....
(1)

WORKING SPACE

Working Unit 06

4MA1-1H Q9

UNIT 1
5 marks**9**Diagram **NOT**
accurately drawn

$ABCD$ is a trapezium with one line of symmetry.

angle $ADC = 60^\circ$ $AD = 12 \text{ cm}$ $DC = 47 \text{ cm}$

Work out the area of the trapezium.
Give your answer correct to 3 significant figures.
Show your working clearly.

..... cm^2 **(Total for Question 9 is 5 marks)**

WORKING SPACE

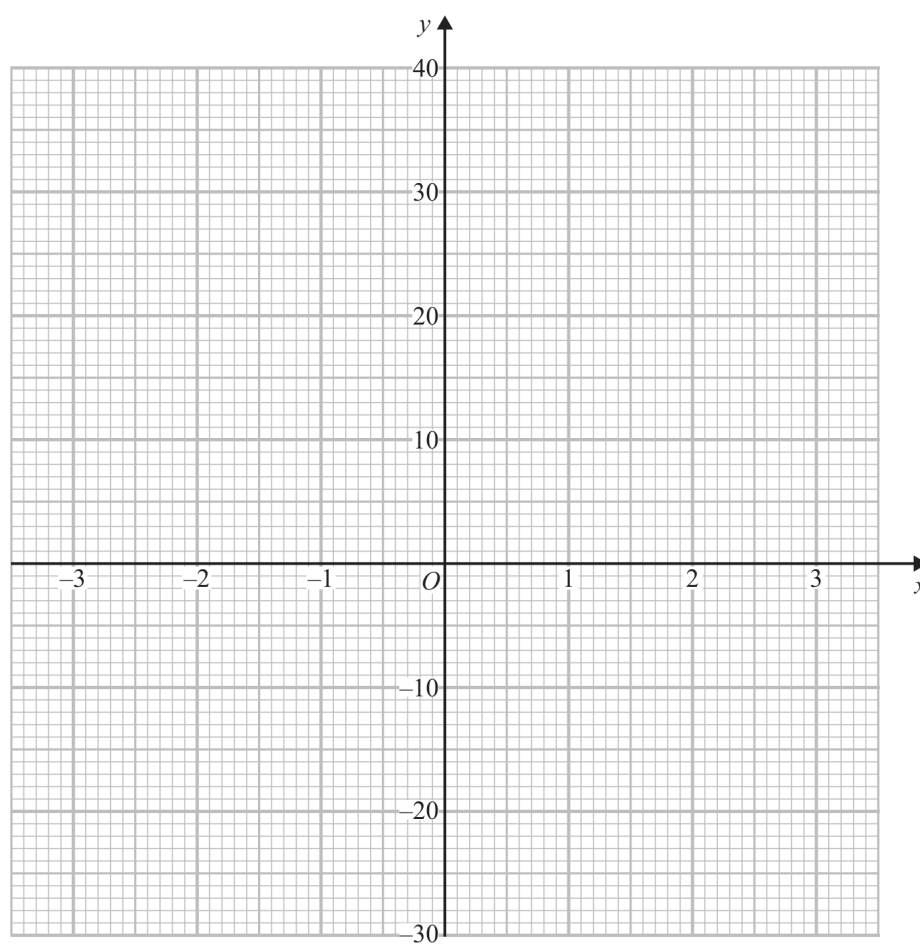
Working Unit 07

4MA1-1H Q10

UNIT 1
3 marks10 (a) Complete the table of values for $y = x^3 + 2x + 3$

x	-3	-2	-1	0	1	2	3
y		-9	0	3	6		36

(1)

(b) On the grid, draw the graph of $y = x^3 + 2x + 3$ for $-3 \leq x \leq 3$ 

(2)

WORKING SPACE

Working Unit 08

4MA1-1H Q11

UNIT 1
2 marks

- 11 $OABC$ is a sector of a circle, centre O

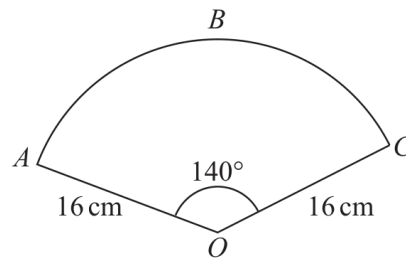


Diagram **NOT**
accurately drawn

$$\text{angle } AOC = 140^\circ$$

$$OA = OC = 16 \text{ cm}$$

Calculate the area of the sector.

Give your answer correct to 3 significant figures.

..... cm^2

(Total for Question 11 is 2 marks)

WORKING SPACE

Working Unit 09

4MA1-1H Q12

UNIT 1
3 marks

12 Express $\frac{5}{4} + \frac{x-3}{6x}$ as a single fraction in its simplest form.

.....
(Total for Question 12 is 3 marks)

WORKING SPACE

Working Unit 10

4MA1-1H Q14

UNIT 1
4 marks

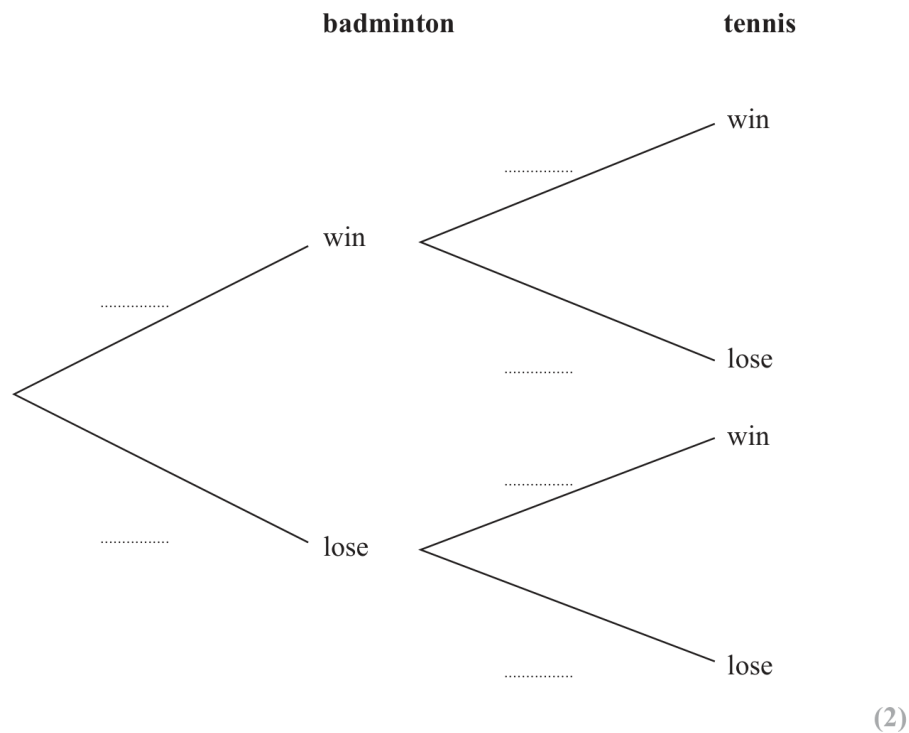
- 14 Ricardo is going to play one game of badminton and one game of tennis.

He can either win or lose each game.

The probability that he will win the game of badminton is 0.7

The probability that he will win the game of tennis is 0.4

- (a) Complete the probability tree diagram.



- (b) Work out the probability that Ricardo loses both games.

.....

(2)

WORKING SPACE

Working Unit 11

4MA1-1H Q15

UNIT 1
4 marks

15 $(\sqrt{3})^5 = k\sqrt{3}$ where k is an integer.

(a) Find the value of k

$$k = \dots\dots\dots$$

(1)

(b) Show that $\frac{21}{3 - \sqrt{2}}$ can be written in the form $c + \sqrt{d}$

where c and d are integers.

Show each stage of your working clearly.

(3)

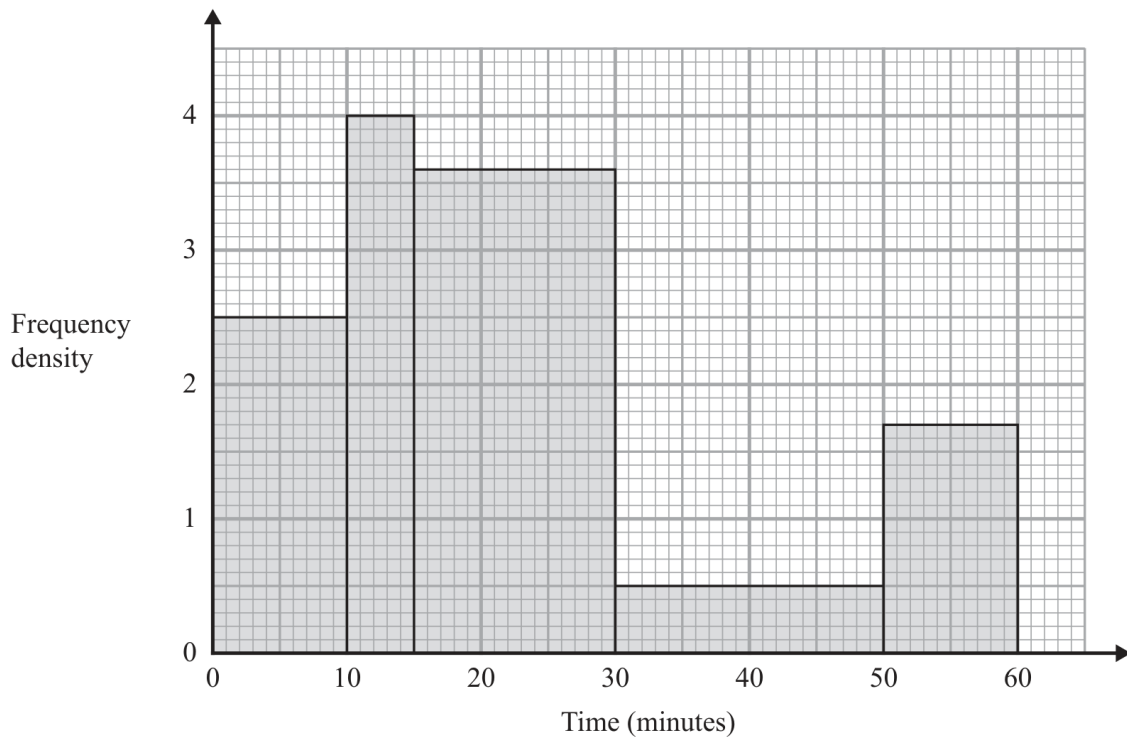
WORKING SPACE

Working Unit 12

4MA1-1H Q17

UNIT 1**3 marks**

- 17 The histogram shows information about the times, in minutes, that some people were in a shop.



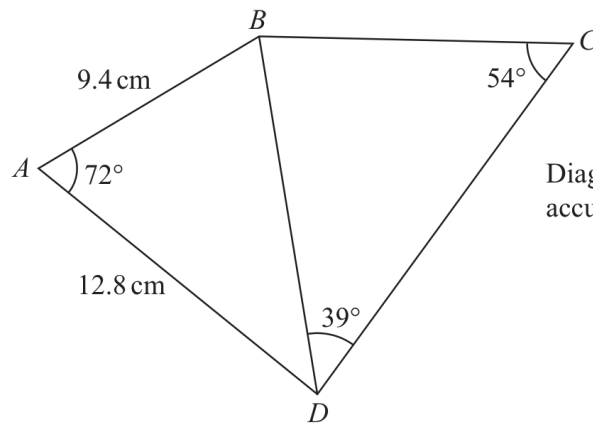
Work out an estimate for the proportion of these people who were in the shop for more than 40 minutes.

(Total for Question 17 is 3 marks)

WORKING SPACE

Working Unit 13

4MA1-1H Q20

UNIT 1
5 marks**20**Diagram **NOT**
accurately drawn

Work out the length of BC
Give your answer correct to 3 significant figures.

..... cm

(Total for Question 20 is 5 marks)

WORKING SPACE

Working Unit 14

4MA1-1H Q21

UNIT 1**3 marks**

21 A box contains 20 counters.

9 of the counters are red

7 of the counters are yellow

4 of the counters are green

Alex takes at random three counters from the box.

Work out the probability that exactly two of the three counters are the same colour.

.....
(Total for Question 21 is 3 marks)

WORKING SPACE

Working Unit 15

4MA1-1H Q25

UNIT 1
5 marks

25 $PQRS$ is a square.

PR is a diagonal of the square.

P is the point with coordinates $(4, 7)$

R is the point with coordinates $(8, -5)$

Find an equation of the straight line that passes through the points Q and S

Give your answer in the form $ay = bx + c$ where a , b and c are integers.

(Total for Question 25 is 5 marks)

WORKING SPACE

Working Unit 16

4MA1-2H Q3

UNIT 1
3 marks

3 Show that $7\frac{1}{3} - 3\frac{4}{7} = 3\frac{16}{21}$

(Total for Question 3 is 3 marks)

WORKING SPACE

Working Unit 17

4MA1-2H Q10

UNIT 1
5 marks

- 10 ABD and ABC are right-angled triangles.

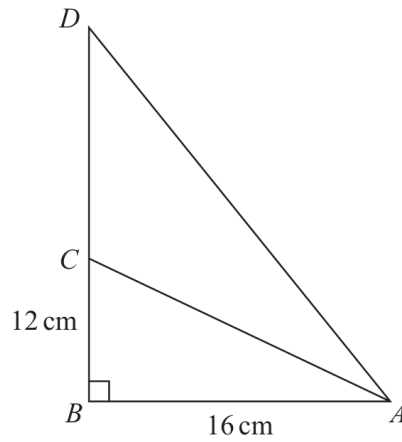


Diagram **NOT**
accurately drawn

$$AB = 16 \text{ cm} \quad BC = 12 \text{ cm} \quad AD = 1.5 \times AC$$

Find the length of CD

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 10 is 5 marks)

WORKING SPACE

Working Unit 18

4MA1-2H Q11

UNIT 1
3 marks

11 Khalid has a box of counters.

17 of the counters are red
28 of the counters are blue
the rest of the counters are orange

Khalid is going to take at random a counter from the box.

The probability that Khalid will take an orange counter is $\frac{4}{9}$

Work out the number of orange counters that are in the box.

(Total for Question 11 is 3 marks)

WORKING SPACE

Working Unit 19

4MA1-2H Q12

UNIT 1
3 marks

12 Expand and simplify $3x(2x+5)(7x-4)$

(Total for Question 12 is 3 marks)

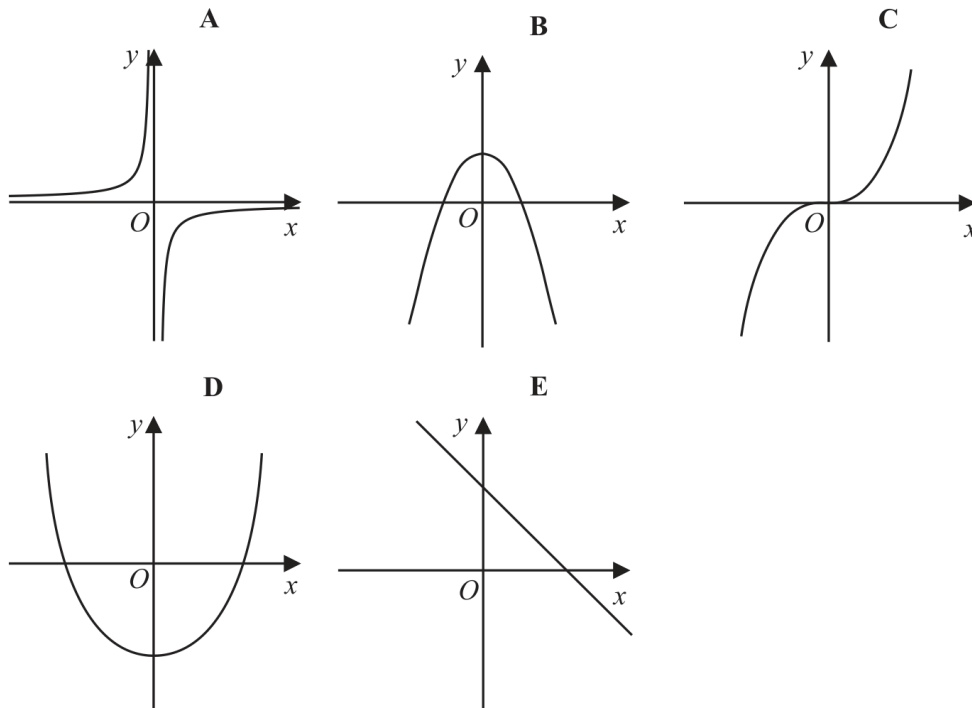
WORKING SPACE

Working Unit 20

4MA1-2H Q14

UNIT 1
2 marks

14 Here are five graphs.

(a) Write down the letter of the graph that could have the equation $y = 5 - x^2$
(1)(b) Write down the letter of the graph that could have the equation $y = 2x^3$
(1)

WORKING SPACE

Working Unit 21

4MA1-2H Q15

UNIT 1
3 marks

15 Simplify fully $\left(\frac{2a^3}{10a^7c^2}\right)^{-3}$

(Total for Question 15 is 3 marks)

WORKING SPACE

Working Unit 22

4MA1-2H Q18

UNIT 1
2 marks

18 Use algebra to show that $0.\dot{7}0\dot{2} = \frac{26}{37}$

(Total for Question 18 is 2 marks)

WORKING SPACE

Working Unit 23

4MA1-2H Q20

UNIT 1
6 marks

20 120 gardeners were asked if they grow carrots (C) or potatoes (P) or tomatoes (T)

Of these gardeners

43 grow carrots

12 grow carrots and potatoes and tomatoes

18 grow carrots and potatoes

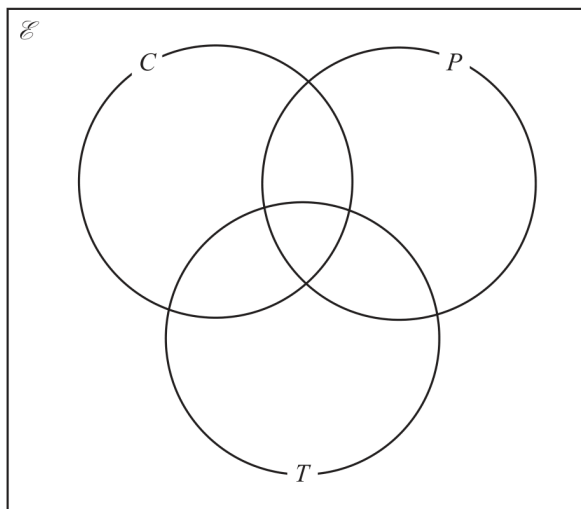
27 grow carrots and tomatoes

32 grow potatoes and tomatoes

29 do not grow carrots or potatoes or tomatoes

The number of these gardeners who grow only potatoes is equal to the number of these gardeners who grow only tomatoes.

(a) Complete the Venn diagram to show this information.



(3)

(b) Find $n(T' \cap C)$

(1)

One of the gardeners who grows carrots is chosen at random.

(c) Calculate the probability that this gardener also grows potatoes.

(2)

WORKING SPACE

Working Unit 24

4MA1-2H Q22

UNIT 1
4 marks

- 22 The diagram shows a square inside rectangle $ABCD$

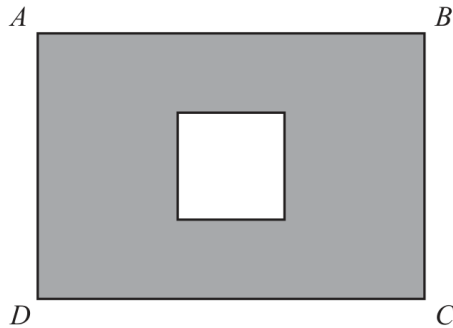


Diagram **NOT**
accurately drawn

The total area of the region shown shaded in the diagram is $X \text{ cm}^2$

$AB = 11.5 \text{ cm}$ correct to the nearest 0.5 cm

$BC = 9.2 \text{ cm}$ correct to 2 significant figures

side of square = 4.1 cm correct to 2 significant figures

By considering bounds, work out the value of X to a suitable degree of accuracy.
Show your working clearly.

$X = \dots\dots\dots$

(Total for Question 22 is 4 marks)

WORKING SPACE

Working Unit 25

4MA1-2H Q23

UNIT 1
4 marks

23 $\frac{4x^2 - 4x - 120}{5x^2 - 180} \div \frac{x^2 + 5x}{10x^2 + 60x} = p$ where p is an integer.

Find the value of p

Show clear algebraic working.

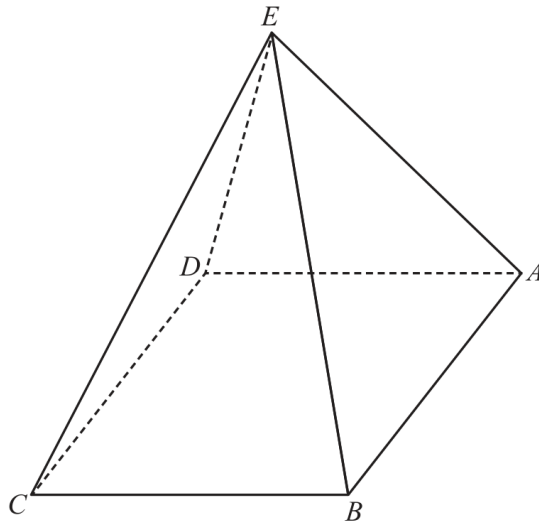
$p = \dots\dots\dots$

(Total for Question 23 is 4 marks)

WORKING SPACE

Working Unit 26

4MA1-2H Q24

UNIT 1
4 marks**24** The diagram shows a square-based pyramid $ABCDE$ Diagram **NOT**
accurately drawn

$$EA = EB = EC = ED$$

 M is the centre of the horizontal square base $ABCD$ Q is the midpoint of AB Angle $EQM = 80^\circ$

$$EA : AB = n : 1$$

Find the value of n

Give your answer correct to 3 significant figures.

 $n = \dots\dots\dots$ **(Total for Question 24 is 4 marks)**

WORKING SPACE

Working Unit 27

4MA1-2H Q25

UNIT 1

3 marks

25 (a) Write $28 + 24x - 6x^2$ in the form $a - b(x - c)^2$ where a , b and c are integers.

.....
(3)

WORKING SPACE

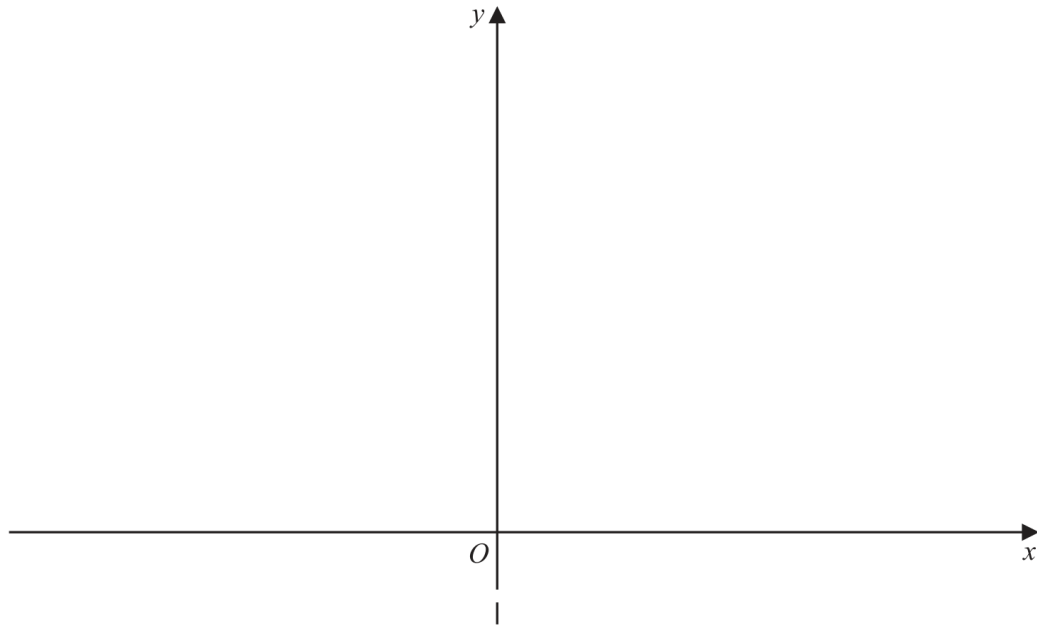
Working Unit 28

4MA1-2H Q25

UNIT 1**3 marks**

(b) On the axes below, sketch the graph of $y = 28 + 24x - 6x^2$

Show clearly the coordinates of the turning point and the coordinates of the point of intersection of the graph with the y -axis.



(3)

WORKING SPACE